

PLACEHOLDER PLATE

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Integrating adaptive grouping into MADDPG

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Many batteries, one street. If each battery thinks only of its own bill, the street still surges. If one office steers them all, the office becomes the bottleneck.

The work is a middle room: agents gather into small groups when the situation asks for it, share what the group needs, and then execute on their own.

On a CityLearn microgrid the public account compares the method with no control, DTDE, CTDE, and a model-predictive bound. Cost -46.6%, peak -37.0%, carbon -38.7%, ramping -22.3% versus no control.